Low Values for Blood Pressure, BMI, and Non-HDL Cholesterol and the Risk of Late-Life Dementia.

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Low values of blood pressure, body mass index (BMI), and non-high-density lipoprotein (HDL) cholesterol have all been associated with increased dementia risk in late life, but whether these risk factors have an additive effect is unknown. This study assessed whether a combination of late-life low values for systolic blood pressure (SBP), BMI, and non-HDL cholesterol is associated with a higher dementia risk than individual low values of these risk factors. This is a post hoc analysis based on an observational extended follow-up of the Prevention of Dementia by Intensive Vascular Care (preDIVA) trial, including community-dwelling individuals, aged 70-78 years and free from dementia at baseline. We assessed the association of baseline low values of SBP, BMI, and non-HDL cholesterol with incident dementia using Cox regression analyses. First, we assessed the respective associations between quintiles of each risk factor and dementia. Second, we explored whether combinations of low values for cardiovascular risk factors increased dementia risk, adjusted for interaction and potential confounders. During a median follow-up of 10.3 years (interquartile range 7.0-10.9 years), 308 of 2,789 participants (11.0%) developed dementia, and 793 (28.4%) died. For all risk factors, the lowest quintile was associated with the highest adjusted risk for dementia. Individuals with 1, 2, and 3 low values had adjusted HRs of 1.18 (95% CI 0.93-1.51), 1.28 (95% CI 0.85-1.93), and 4.02 (95% CI 2.04-7.93), respectively, compared with those without any low values. This effect was not driven by any specific combination of 2 risk factors and could not be explained by competing risk of death. Older individuals with low values for SBP, BMI, or non-HDL cholesterol have a higher dementia risk compared with individuals without any low values. Dementia risk was substantially higher in individuals with low values for all 3 risk factors than expected based on a dose-response relationship. This suggests the presence of an overarching phenomenon that involves multiple risk factors simultaneously, rather than resulting from independent effects of each individual risk factor. ISRCTN registry preDIVA: ISRCTN29711771. Date of study submission to ISRCTN registry: February 14, 2006. Recruitment start date: January 1, 2006. doi.org/10.1186/ISRCTN29711771.

Cognitive decline precedes late-life longitudinal changes in vascular risk factors.

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Although obesity, hypercholesterolaemia and hypertension in midlife are risk factors for dementia in late life, dementia is associated with lower body mass index, cholesterol levels and blood pressures. It is unclear whether declines in these vascular risk factors are preceded by declines in cognitive function or vice versa. Within the Leiden 85-plus Study, a prospective population-based study of 599 subjects aged 85 years, the authors annually measured body mass index, total cholesterol, high-density lipoprotein (HDL) cholesterol, glucose levels and blood pressure, and assessed global cognitive function using the Mini Mental State Examination (MMSE) during a 5-year follow-up. For the whole population who survived up to the age of 90 years, strong annual declines in MMSE score, body mass index, total cholesterol levels, glucose levels, and blood pressure, and an annual increase in HDL cholesterol levels were observed during the follow-up period (all p< or =0.010). Annual changes in MMSE score from age 85 to 87 years were associated positively with annual changes from age 87 to 90 years in total and HDL cholesterol levels (p=0.002 and p=0.013), systolic and diastolic blood pressure (p=0.008 and p=0.048), but not BMI. Parameter value changes from age 85 to 87 years were not associated with changes in MMSE score from age 87 to 90 years. In old age, cognitive decline precedes declines in total cholesterol levels, HDL cholesterol levels and blood pressure, and not vice versa. Possibly, brain lesions in metabolic and blood pressure regulation centres cause dysregulation of lipid metabolism and blood pressure.

Evaluation of the Concurrent Trajectories of Cardiometabolic Risk Factors in the 14 Years Before Dementia.

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Cardiometabolic risk factors have been associated with an increased risk of dementia; yet, the optimal targets and time window for the management of cardiometabolic health to prevent dementia remain unknown. To model concurrently and compare the trajectories of cardiometabolic risk factors up to 14 years preceding diagnosis in individuals with dementia and matched controls free of dementia. A case-control study nested within the Three-City study, a French population-based cohort of older persons (≥65 years), included 6 home visits with neuropsychological testing between 1999 and 2014. Data analysis was performed in September 2017. A total of 785 incident dementia cases and 3140 controls matched by sex, age, educational level, and cohort center at the time of diagnosis were evaluated. Repeated measures of body mass index (BMI) and systolic (SBP) and diastolic (DBP) blood pressure, high-density lipoprotein (HDL-C) and low-density lipoprotein cholesterol (LDL-C), triglycerides, and glycemia levels between 1999 and 2014. Incidence of dementia based on systematic detection and validated diagnosis. A total of 785 cases and 3140 controls (2530 [65%] women; mean [SD] age, 76 [5] years) were included in the study. Cases presented a faster decline in BMI, slower increase of SBP and constantly lower DBP. Mean values (95% CI) 14 years before diagnosis (-14 years) and at diagnosis (year 0) for the most common profile were BMI, 26.1 (25.6-26.5) and 24.8 (24.5-25.1) for a case, and 25.7 (25.4-26.1) and 25.3 (25.0-25.5) for a control; for SBP, 135.2 (131.8-138.7) and 142.1 (140.3-143.9) mm Hg for a case, and 135.8 (132.9-138.6) and 144.9 (143.7-146.1) mm Hg for a control; for DBP, 76.5 (74.7-78.5) and 74.0 (73.1-74.9) mm Hg for a case, and 76.7 (75.1-78.3) and 75.0 (74.2-75.7) mm Hg for a control. In contrast, glycemia was higher among cases (mean fasting glucose values [95% CI] at -14 years and year 0: 89.4 [86.9-92.1] and 96.4 [93.7-99.3] mg/dL for a case, and 87.1 [85.1-89.2] and 95.3 [93.5-97.1] mg/dL for a control), with a significant case-control difference from -1.6 to -14 years prior to diagnosis. There were no significant case-control differences in trajectories of blood lipid levels (mean values [95% CI] at -14 years and year 0: for HDL-C, 70.6 [67.6-73.9] and 61.3 [58.9-63.8] mg/dL for a case, and 70.4 [67.5-73.3] and 62.3 [60.2-64.3] mg/dL for a control; for LDL-C: 147.2 [140.5-154.5] and 141.6 [136.6-146.7] mg/dL for a case, and 144.3 [138.7-150.4] and 141.2 [137.5-145.2] mg/dL for a control; for triglycerides: 115.5 [103.6-149.1] and 112.6 [104.8-120.9] mg/dL for a case, and 112.5 [103.8-144.4] and 109.7 [105.0-114.8] mg/dL for a control). In this large cohort of older persons, BMI declined in prodromal dementia, possibly reflecting early preclinical changes. Lower BP prior to dementia may reflect both a consequence and a contributing factor for the disease, whereas higher blood glucose levels may constitute a risk factor for dementia in the older age range. Overall, these findings suggest that elevated glycemia, low BP, and weight loss may be primary targets for the management of cardiometabolic health for primary and secondary prevention of dementia in the older age range.

Adverse Lipid Profiles Are Associated with Lower Dementia Risk in Older People.

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Midlife dyslipidemia is associated with higher risk of dementia in late-life dementia, but the impact of late-life dyslipidemia on dementia risk is uncertain. This may be due to the large heterogeneity in cholesterol measures and study designs employed. We used detailed data from a large prospective cohort of older persons to comprehensively assess the relation between a broad range of cholesterol measures and incident dementia, addressing potential biases, confounders, and modifiers. Post hoc observational analysis based on data from a dementia prevention trial (PreDIVA). 3392 community-dwelling individuals, without dementia, aged 70-78 years at baseline (recruited between June 2006 and March 2009). Total cholesterol, low-density lipoprotein cholesterol (LDL-C), high-density lipoprotein cholesterol (HDL-C), triglycerides, and apolipoprotein A1 and B were assessed. Over a median of 6.7 years' follow-up, dementia was established by clinical diagnosis confirmed by independent outcome adjudication. Hazard ratios (HRs) for dementia and mortality were calculated using Cox regression. Dementia occurred in 231 (7%) participants. One-SD increase in LDL/HDL conveyed a 19% ($P = .01$) lower dementia risk and a 10% ($P = .02$) lower risk of dementia/mortality combined. This was independent of age, cardiovascular risk factors, cognitive function, apolipoprotein E genotype, and cholesterol-lowering drugs (CLD). This association was not influenced by the competing risk of mortality. Consistent and significant interactions suggested these associations were predominant in individuals with low body mass index (BMI) and higher education. Dyslipidemia in older individuals was associated with a lower risk of dementia. Low BMI and higher education level mitigate poor outcomes associated with dyslipidemia. These findings suggest that a different approach may be appropriate for interpreting lipid profiles that are conventionally considered adverse in older adults. Such an approach may aid predicting dementia risk and designing intervention studies aimed at reducing dementia risk in older populations.

Fontan circulation and systemic disease - a retrospective cohort analysis over 35 years of follow-up.

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The Fontan operation provides lifesaving palliation for individuals with single ventricle (SV) physiology. Given recent concerns of systemic disease (SD) for patients with a Fontan circulation, we sought to (1) quantify the increase in SD incidence associated with the Fontan circulation; (2) identify the risk factor of SD; (3) assess the association between SD and mortality in patients with a Fontan circulation. A matched retrospective cohort study design was adopted. From the Quebec Congenital Heart Disease (CHD) Database with up to 35 years of follow-up, patients who survived at least 30 days after the Fontan operation were identified. For each Fontan patient, patients with isolated ventricular septal defect (VSD) with the same sex and age were identified and 20 of them were randomly selected to form the control group. The presence of SD was defined as at least 1 hospitalization due to extra-cardiac complications including liver, respiratory, gastrointestinal or renal disease. Time-to-event analysis including Kaplan-Meier curve analysis and Cox proportional hazard models were conducted to assess the cumulative risk of SD, risk factors of SD, and the association between SD and 10-year mortality. A total of 533 patients with Fontan circulation were identified and matched with 10,280 VSD patients. The cumulative probabilities of SD at 10- and 35-years follow-up were 59.02% and 89.66% in patients with a Fontan circulation, 4 to 7 times of the probabilities in VSD patients (8.68% and 23.34%, respectively; LogRank tests P 1 extra-cardiac organ affected) disease was associated with a 3.38-fold (95%CI: 1.73-6.60) increase in 10-year mortality risk when comparing to the absence of SD. This population-based study demonstrated that patients with a Fontan circulation had increased risk of SD, which in turn led to higher risk of mortality. These findings underscore the need for more systematic surveillance of cardiac and systemic disease for patients after Fontan operation.

Functional and imaging outcomes of the Fontan circulation following pregnancy.

PMID: 39547425 | Indexed in PubMed

The Fontan circulation palliates single-ventricle congenital heart disease by separating the systemic and pulmonary circulations. An increasing number of women with a Fontan circulation are wishing to become pregnant, however the ability to increase cardiac output during pregnancy is limited in many due to the chronic low output state. We describe pregnancy outcomes in these women at a large tertiary centre, including functional and imaging outcomes. We retrospectively investigated women with a Fontan circulation giving birth after 24 weeks' gestation between 1995 and 2023. Data collected included obstetric and neonatal complications, changes in cardiac volumes and ejection fraction (EF), and functional outcomes including change in NYHA class and exercise capacity pre- and post-pregnancy, compared to matched male controls. Twenty-six pregnancies occurred among 23 women. Almost half experienced obstetric complications, primarily bleeding, which was the commonest indication for emergency C-section. Worsening cardiac symptoms complicated 50 % of pregnancies, with 4 requiring hospital admission for decompensation. Arrhythmias were not uncommon. Mean VO2max declined post-pregnancy ($p = 0.03$), though not significantly compared to controls. Worsening of NYHA class was uncommon (15.4 %), suggesting that cardiovascular complications during pregnancy do not correlate with longer-term functional limitation. EF worsened post-pregnancy, declining significantly more so than in matched controls ($p = 0.03$), however there were no changes in cardiac volumes. Pregnancy in women with a Fontan circulation is associated with high rates of obstetric complications and worsening cardiac symptoms. However, successful pregnancies are possible, with little evidence of medium-term functional impairment. This data informs pre-pregnancy counselling of Fontan women of child-bearing age.

Anaesthetic management of a child with Fontan heart undergoing laparoscopic cholecystectomy: A case report.

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The Fontan procedure is a staged palliative surgery performed in children with congenital univentricular heart defects. These individuals are predisposed to a variety of issues due to their altered physiology. Through this article, we would like to describe the evaluation and anaesthetic management of a 14-year-old boy with Fontan circulation who underwent an uneventful laparoscopic cholecystectomy. The key to successful management was a multidisciplinary approach throughout the perioperative period as these patients pose a unique set of problems.

Five decades of Fontan palliation: What have we learned? What should we expect?

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The Fontan procedure is the final palliative surgery in a series of staged surgeries to reroute the systemic venous blood flow directly to the lungs, with the ventricle(s) pumping oxygenated blood to the body. Advances in medical and surgical techniques have improved patients' overall survival after the Fontan procedure. However, Fontan-associated chronic comorbidities are common. In addition to chronic cardiac dysfunction and arrhythmias, complications involving other organs such as the liver, lungs, intestine, lymphatic system, brain, and blood frequently occur. This narrative review focuses on the immediate and late consequences in children, pregnant women, and other adults with Fontan circulation. In addition, we describe the technical advancements that might change the way single-ventricle patients are managed in future.

Four decades of Fontan palliation.

PMID: 20585329 | Indexed in PubMed

The Fontan palliation was introduced in 1968 to treat cardiac malformations unsuitable for biventricular repair. This procedure has transformed the surgical management of congenital heart disease. In this Review, we reflect on the outcomes and clinical problems associated with this unique circulation after more than 40 years of experience. We also summarize the evolution of the Fontan procedure, highlight the long-term clinical issues and their management, and consider future expectations of a circulation driven by a single ventricle with the systemic and pulmonary blood flow in series rather than in parallel.

Considerations in Critical-Care and Anesthetic Management of Adult Patients Living With Fontan Circulation.

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The Fontan procedure is a staged palliation for various complex congenital cardiac lesions, including tricuspid atresia, pulmonary atresia, hypoplastic left heart syndrome, and double-inlet left ventricle, all of which involve a functional single-ventricle physiology. The complexity of the patients' original anatomy combined with the anatomic and physiologic consequences of the Fontan circulation creates challenges. Teens and adults living with Fontan palliation will need perioperative support for noncardiac surgery, peripartum management for labour and delivery, interventions related to their structural heart disease, electrophysiology procedures, pacemakers, cardioversions, cardiac surgery, transplantation, and advanced mechanical support. This review focuses on the anesthetic and intensive care unit (ICU) management of these patients during their perioperative journey, with an emphasis on the continuity of preintervention planning, referral pathways, and postintervention ICU management. Requests for recipes and doses of medications are frequent; however, as in normal anesthesia and ICU practice, the method of anesthesia and dosing are dependent on the presenting medical/surgical conditions and the underlying anatomy and physiologic reserve. A patient with Fontan palliation in their early 20s attending school full-time with a cavopulmonary connection is likely to have more reserve than a patient in their late 40s with an atriopulmonary Fontan at home waiting for a heart transplant. Each case will require an anesthetic and critical care plan tailored to the situation. The critical care environment is a natural extension of the anesthetic management of a patient, with complex considerations for a patient with Fontan palliation.